Unit 2

Campbell Biology, AP Edition, 12e

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The Cell

Dr. Diana Bautista

Dr. Diana Bautista is an Associate Professor of Molecular and Cell Biology at the University of California, Berkeley, and a Howard Hughes Medical Institute Faculty Scholar. Recipient of the Gill Transformative Investigator Award and the Society for Neuroscience Young Investigator Award, she received a B.S. in Biology from the University of Oregon and a Ph.D. in Neuroscience from Stanford University. Dr. Bautista grew up in Chicago, the first in her family to attend college and to pursue a career in science. She and her laboratory members investigate the cell-signaling pathways associated with pain and itch and how their

misregulation can result in chronic pain and chronic itch. Her research is interdisciplinary: Dr. Bautista collaborates with computational biologists, immunologists, and physiologists. She believes that science benefits most from a diverse group of colleagues, each bringing a different perspective.

An Interview with: Diana M. Bautista

What got you interested in biology, and in neuroscience in particular?

I always liked science as a kid, but it was never a big focus in my life. I went to college interested in art, then took some time off because I was a terrible artist. I ended up working with an environmental group helping low-income communities oppose a hazardous waste incinerator. That got me excited about science because it was at the interface of the environment and community empowerment, so I decided to return to college and major in biology. Once I returned to school, at the University of Oregon, I found out you could get a work-study job in a lab. I ended up working in Peter O'Day's lab studying vision in fruit flies by recording electrical signals in response to light. For me, shining light on a fly eye and recording an electrical signal from their nervous system in real time was just the coolest thing I had ever seen. The lab was a really inclusive, warm environment, and Peter was an amazing mentor—if it weren't for him, I wouldn't be in science. I went to graduate school at Stanford in neuroscience, where I worked with Dr. Richard Lewis on a newly discovered calcium channel and how calcium signaling pathways drive cellular behaviors.

Why did you focus your research on pain?

After my Ph.D., I wanted to continue working on cell signaling but from a more organismal viewpoint, so I did my post-doctoral research with Dr. David Julius at the University of California, San Francisco. His lab had recently identified the cellular protein (TRPV1) that is responsive to capsaicin—the molecule that gives chili peppers their "hotness"—and showed that this protein is also activated by painful heat. That was really exciting to me because it involved thinking about proteins involved in cell signaling, but in the bigger context of pain. Proteins like TRPV1 serve important protective functions—for example, by stopping us from grabbing a hot frying pan that could cause a burn. One of my projects was to identify the wasabi receptor, a protein on the surface of nerve cells that plays a key role in inflammatory pain hypersensitivity. Once you activate this receptor protein—with irritants such as wasabi, mustard oil, or inflammatory mediators produced in the body by tissue injury such as a burn—it activates and makes pain cells more sensitive: Warmth triggers sensations of burning heat, and light touch triggers pain. Normally, pain hypersensitivity dies down after a few days as the tissue heals, but in some patients, it becomes a chronic debilitating disease.

What does your lab work on now?

We are interested in the sense of touch. Touch-sensitive neurons innervate the skin and mediate gentle touch sensations and texture preferences: Do you like wool sweaters, or do you find them itchy? Touch-sensitive neurons also innervate our musculature, allowing us to detect limb position and mediating coordinated movements, such as our ability to text or to walk in a straight line without staring at our feet. In someone suffering from chronic pain, a gentle touch feels incredibly painful, while a chronic itch sufferer experiences light touch as itchy and pain as pleasure. When you scratch an itch, you're scratching really hard—if you didn't have an itch it would hurt, but when you do have an itch it feels good. We have no idea how our body processes the same type of stimulus under different injury or disease conditions, and that's what really excited me when I started my own lab.

Dr. Bautista studies cell responses to stimuli using equipment that measures electrical signals.

What is your advice to an undergraduate considering a career in biology?

I'm an advisor for undergraduates, so I talk to a lot of students. I like to tell them about my path and let them know that it doesn't matter what you did before or where you come from—anybody can be a scientist. I think it's sometimes tough early on in biology, because at first you're taught facts, but as you progress, biology is

about figuring out the unknown. And if you can get into a lab early on, then you really see that side of science—that it's about learning what we know, and then going beyond that, doing experiments to identify new mechanisms.

> ***"It doesn't matter what you did before or where you come from—anybody can be a scientist."***